The University of Azad Jammu and Kashmir Muzaffarabad



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Digital Image Processing _Lab

5th Semester

2024-28 (SE)

Lab_01

Department Of Software Engineering

Step 1: Python installation



Figure 1 Python installation



Figure 2 installing



Figure 3 installation successful

Step 2: Python and pip verification

python --version

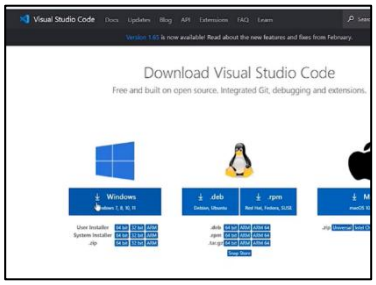
python -m pip --version

```
C:\Users\jawah_czbwkdr>python --version
Python 3.14.8

C:\Users\jawah_czbwkdr>python -m pip --version
pip 26.2.1 from C:\Users\jawah_czbwkdr\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\pip (python 3.14)
```

Figure 4 Python and pip verification

Step 3: Install VS Code and Select Python



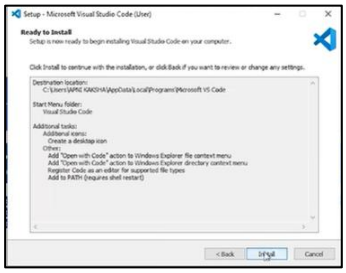


Figure 5 VS Code Installation

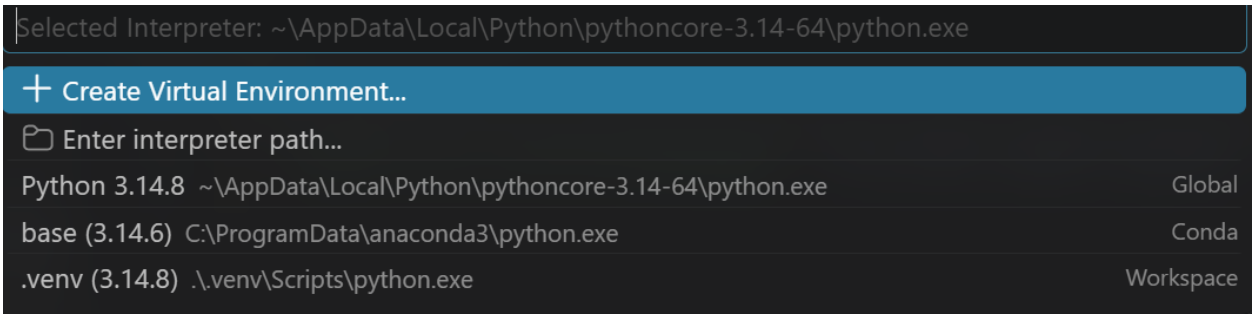
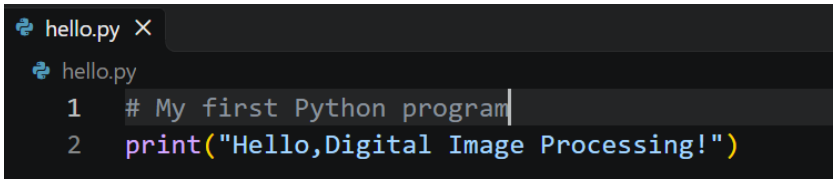


Figure 6 Select Interpreter

Step 4: Run Your First Python Program

Code:



Output:

```
PS C:\Users\jawah_czbwkdr\dip-lab-01> & C:\Users\jawah_czbwkdr\AppData\Local\Python\pythoncore-3.14-64\python.exe c:/Users/jawah_czbwkdr/dip-lab-01/hello.py
Hello,Digital Image Processing!
```

Step 5: Basic Python Exercises

Exercise 1: Variables and Data Types

Code:

```python
# Exercise 1: Variables and Data Types
width = 640
height = 480
image_name = "sample.png"
is_color = True
print(width, type(width))
print(height, type(height))
print(image_name, type(image_name))
print(is_color, type(is_color))
print("Total pixels:", width * height)   # 307200
```

Output:

```
640 <class 'int'>
480 <class 'int'>
sample.png <class 'str'>
True <class 'bool'>
Total pixels: 307200
```

Exercise 2: if, elif, and else

Code:

```python
# Exercise 2: if, elif, else
for intensity in [-1, 0, 127, 128, 255, 256]:
    if intensity < 0 or intensity > 255:
        label = "Invalid"
    elif intensity >= 128:
        label = "Bright pixel"
    else:
        label = "Dark pixel"
    print(intensity, "->", label)
```

Output:

```
-1 -> Invalid
0 -> Dark pixel
```

Exercise 3: for and while Loops

Code:

```python
# Exercise 3: for and while loops
for value in [20, 130, 255]:
    if value >= 128:
        print(value, "Bright pixel")
    else:
        print(value, "Dark pixel")

counter = 0
while counter < 3:
    print(counter)
    counter += 1
```

Output:

```
20 Dark pixel
130 Bright pixel
255 Bright pixel
0
1
2
```

Exercise 4: Functions

python basics.py > ...

```python
# Exercise 4: Functions
def is_bright(pixel, threshold=128):
    return pixel >= threshold

print(is_bright(127))        # False
print(is_bright(128))        # True
print(is_bright(200, 210))   # False
```

Output:

```
PS C:\Users\jawah_czbwkdr\dip-lab-01> & C:\Users\jawah_czbwkdr\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/jawah_czbwkdr/dip-lab-01/python basics.py"
False
True
False
```

Exercise 5: Lists and Tuples

Code:

```
python basics.py > ...
  4  print(pixels[0])
  5  pixels[1] = 150
  6  pixels.append(64)
  7  print(pixels, len(pixels))   # [0, 150, 255, 64] 4
  8
  9  # A tuple is immutable, so its items cannot be changed once created.
 10  size = (640, 480)
 11  width, height = size
 12  print(width, height)
 13
 14
```

```
2↳ print("Hello, Digital Image Processing!"
hello.py                                Tab to open →
```

Output:

```
PS C:\Users\jawah_czbwkdr\dip-lab-01> & C:\Users\jawah_czbwkdr\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/jawah_czbwkdr/dip-lab-01/python basics.py"
0
[0, 150, 255, 64] 4
640 480
PS C:\Users\jawah_czbwkdr\dip-lab-01>
```

Exercise 6: Dictionaries and Sets

hello.py | python basics.py ×

```
python basics.py > ...
  1  # Exercise 6: Dictionaries and Sets
  2  image = {"width": 640, "height": 480}
  3  image["mode"] = "RGB"
  4  print(image["width"], image["mode"])
  5
  6  labels = set(["cat", "dog", "cat", "bird"])
  7  print(len(labels))   # 3
```

Output:

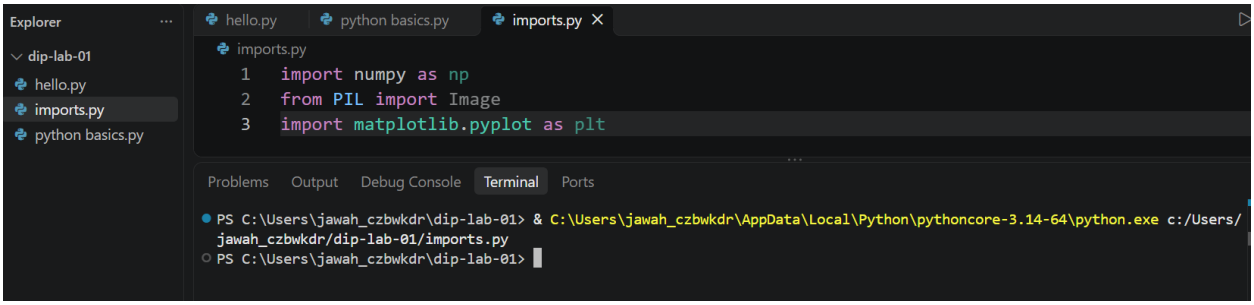
```
PS C:\Users\jawah_czbwkdr\dip-lab-01> & C:\Users\jawah_czbwkdr\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/jawah_czbwkdr/dip-lab-01/python basics.py"
640 RGB
3
PS C:\Users\jawah_czbwkdr\dip-lab-01>
```

Step 6: Install the Required Packages

- python -m pip install numpy pillow matplotlib
- python -m pip show numpy pillow matplotlib

```
PS C:\Users\jawah_czbwkdr\dip-lab-01> python -m pip list
contourpy       1.4.0
cycler          0.12.1
fonttools       4.67.0
kiwisolver      1.5.1
matplotlib      3.11.2
numpy           2.5.4
packaging       26.3
pillow          13.0.0
pip             26.2.1
pyparsing       3.3.3
python-dateutil 2.9.0.post0
six             1.17.0
PS C:\Users\jawah_czbwkdr\dip-lab-01>
```

Step 7: lab01_imports.py



Python versus pip and the purpose of the import statements

Python is the programming language used to write and run programs, while pip is its package manager used to download and install external libraries like NumPy, Pillow and Matplotlib. Import statements load these libraries into a program so their functions can be used in the code.